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 Studierichting: Informatica
 Oefeningenreeks 6A nr. 8.8

$$x_6 = ?? \text{ voor } \left(\frac{3}{4}, \frac{1}{4}, \frac{1}{12}, \dots \right)$$

$$\begin{cases} \frac{1}{4} - \frac{3}{4} = -\frac{1}{2} \\ \frac{1}{12} - \frac{1}{4} = -\frac{1}{6} \end{cases}$$

$\Rightarrow$ geen rekenkundige rij

$$\begin{cases} r = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3} \\ r = \frac{\frac{1}{12}}{\frac{1}{4}} = \frac{1}{3} \end{cases}$$

$\Rightarrow$ meetkundige rij

$$x_n = x_1 \cdot r^{n-1}$$

$$x_n = \frac{3}{4} \cdot \left(\frac{1}{3} \right)^{n-1}$$

$$x_6 = \frac{3}{4} \cdot \left(\frac{1}{3} \right)^5$$

$$x_6 = \frac{1}{324}$$

References